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Seizure detection using EEG on the CHB-MIT dataset via multi-domain feature engineering and classical machine learning

Srishti Ghosh[1], Vandita Singh[1] and Omkar S Powar[1*]

*Correspondence:
Omkar S Powar
omkar.powar@manipal.edu
[1]Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal 576104, Karnataka, India

Abstract

Seizure detection from scalp EEG recordings requires features capable of capturing the rapidly evolving patterns of cerebral activity. We present a strictly patient-independent approach that investigates 1-second EEG windows using nested cross-validation to prevent data leakage within the paediatric CHB-MIT dataset. In this approach, we combine features extracted from temporal, spectral, wavelet, and spatial (CSP) domains. Feature selection, spatial filtering, and hyperparameter optimization are performed only within training folds to avoid leakage. A compact feature set of just 15 features enables traditional classifiers to achieve remarkable performance: a K-Nearest Neighbours classifier achieves 90.6% accuracy, whereas a Random Forest achieves 90.5% accuracy with an AUC of 0.927. The proposed system is interpretable, highly computationally efficient, and suitable for near real-time clinical monitoring.

Keywords EEG, Seizure detection, CHB-MIT, Feature engineering, Machine learning, Random forest, Common spatial patterns, Wavelet transform

1 Introduction

The human brain, comprising billions of neurons communicating through electrical signals, produces a complex pattern we can measure with EEG. Invented by Hans Berger, EEG can record these signals non-invasively via scalp electrodes. EEG is used commonly in the investigation of epilepsy, Alzheimer's disease, and sleep disorders. However, it is difficult to interpret because brain activity always keeps fluctuating and might, therefore, be buried under noise [1].

Epilepsy affects approximately 70 million people worldwide. Recurring seizures, often with no warning, are caused by sudden bursts of abnormal electrical activity in the brain. Such unpredictable episodes cause injuries, emotional distress, and, on occasions, life-threatening situations. In many parts of the world, especially where neurological specialists are scarce, early detection is imperative to ensure patient safety and a good quality of life. Currently, physicians manually go through hours of EEG recordings. This is time-consuming and, to an extent, subjective. It has been reported in some studies that as

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many as 30% of seizures are misdiagnosed. This has driven the development of automated seizure detection systems that can support clinicians [2].

We used the CHB-MIT Scalp EEG Database, which includes recordings of paediatric patients. We standardized the data on a consistent sampling rate of 256 Hz, considering only 1-second segments of brain activity over 23 electrode channels. Segments in this dataset are clearly annotated as seizure or non-seizure, providing a strong foundation for training detection algorithms [3, 4].

Feature extraction transforms raw EEG signals into meaningful numerical representations that machine learning algorithms can understand. Since the nature of EEG signals is highly temporal, we extracted features from three domains: time, containing signal shape information; frequency, containing rhythmic patterns; and time-frequency, containing transient details. For all these domains, we computed several statistical measures related to mean, variance, and indicators of complexity. Afterward, the most discriminating features were selected for building an efficient system [5, 6].

Classic machine learning methodologies such as Support Vector Machines and Random Forests remain highly useful due to their transparency, efficiency, and reliability when combined with a strong feature set. In clinical environments, interpretability and low latency are prerequisites for trustworthiness and for facilitating timely interventions, hence a feature-based approach is particularly fitting for real-time decision support [7].

The objective is to build an accurate and interpretable seizure-detection system by utilizing multi-domain feature extraction in collaboration with classical machine learning. The aim is to achieve high performance while maintaining computational efficiency compatible with clinical deployment. The proposed system analyses 1-second windows of EEG data, thus holding the potential to support real-time alerts in healthcare. This paper is organized as follows: Sect. II discusses the related work, Sect. III describes the methodology followed, Sect. IV presents the results, and Sect. V offers concluding remarks and discusses directions for future work.

2 Related work

Recent studies on automated seizure detection have diverged into two main directions: classic machine learning using hand-designed features and pure end-to-end deep learning models. Numerous studies have utilized publicly available datasets like CHB-MIT, enabling comparative analysis [8].

Traditional approaches typically segment EEG data and extract features from the time, frequency, and wavelet domains, respectively. Then classify these with methods like support vector machines or Random Forests. Spatial analysis methods-also known as spatial filtering-such as Common Spatial Patterns (CSP), improve detectability by highlighting seizure-related activity at different electrode sites [9]. Modern feature engineering work, such as NeuroFeat in 2025, continues to support the idea that effective, interpretable seizure detection can be achieved with features drawn from the data. Classical methods remain highly competitive, particularly when analysis is strictly patient-independent and free from data leakage [10].

In contrast, deep learning approaches, such as CNNs or hybrid architectures, usually achieve high sensitivity but demand large amounts of data, suffer from issues regarding interpretability, and sometimes have low generalizability across recording conditions [11, 12]. Note that the accuracies reported by different studies are not comparable because of

heterogeneous evaluation protocols; for example, the most common patient-dependent versus patient-independent splits, overlapping windows, and feature selection without leakage. Correspondingly, this work focuses on patient-independent evaluation to best reflect possible clinical deployment scenarios.

A summary comparison of contemporary methodologies, highlighting feature engineering and validation rigor, is presented in Table 1. While the deep learning approaches yield high performance and robust feature-based methods set benchmarks for clinical wearability [10], recent strides have underlined the effectiveness of advanced, interpretable feature construction [23, 24]. The present work combines the multi-domain feature innovation typical of this latter paradigm with the rigorous patient-independent validation demanded by clinical deployment.

3 Methodology

In this section, the proposed seizure detection approach using EEG data is described in a step-by-step manner, starting from data preparation to classification.

3.1 Overview of the pipeline

Our methodology prioritizes both accuracy and interpretability through a structured pipeline. The continuous recordings are preprocessed and segmented into

Table 1 Comparative analysis of EEG seizure detection methods on CHB-MIT and related datasets

Study	Dataset & cohort	Features/model	Key metrics	Evaluation protocol & clinical relevance
Acharya et al. (2012) [13] (classic features)	Bonn EEG (curated segments)	Entropy, PCA + Wavelet + Classifiers (e.g., SVM, NN)	Accuracy > 97%	Highly curated, single-patient segments. Demonstrates the power of good features but on a simplified, non-continuous task
Omidvar et al. (2021) [14] (wavelet + DWT feature eng.)	Bonn EEG (curated segments)	5-level DWT + GA-based feature selection + ANN/SVM	Reported high accuracy	Method-focused on feature selection. Highlights the importance of domain-specific feature engineering, though on a simpler dataset
Ingolfsson et al/ (2024) [10] (robust clinical approach)	CHB-MIT (wearable context)	Feature-based Gradient Boosted Trees with dedicated artifact handling	Sensitivity 65.27%, low FP/hr	Patient-independent, prioritizes near-zero false alarms (clinical necessity). A key benchmark for realistic, wearable application
Das et al. (2025) [15] (Innovative geometric feature engineering)	Prenatal EEG (Meditative States)	Novel feature representation: Tetrahedral feature maps for EEG geometry + ML classifiers	Accurate classification	Demonstrates innovation in feature space construction. Shows the field's move beyond simple statistics to geometric and structural feature engineering, directly relevant to your multi-domain approach
Choudhury et al. (2026) [16] (State-of-the-art adaptive feature engineering)	Multi-centre MDD EEG	Adaptive feature engineering: framework for automatic, adaptive feature learning from raw EEG	AUC ~ 94% (for MDD)	Interpretable, patient-independent. This is a crucial citation as it represents the current pinnacle of interpretable, engineered feature research, which is the paradigm your work follows
Mekruksavanich et al. (2024) [17] (Representative modern deep learning)	CHB-MIT	Enhanced Hybrid CNN with Attention Mechanism (end-to-end)	High Sensitivity/AUC (e.g., > 99% CV)	High Sensitivity/AUC (e.g., > 99% CV)
Proposed study	**CHB-MIT paediatric** [3]	**Multi-domain features + MI** → SFS ($n = 15$)	**RF: Acc 90.5% ± 1.2%, AUC 0.927 ± 0.015**	**Interpretable, low latency**

non-overlapping 1-second windows (256 samples per channel at 256 Hz). It then extracts features part by part across multiple domains like time, frequency, wavelet, along with spatial patterns. In the next phase, feature selection is done to keep only the most informative ones. This is followed by classification with one of KNN, SVM, or Random Forest. Finally, performance evaluation will be obtained through various metrics and statistical tests. The overall pipeline is illustrated in Fig. 1.

3.2 Dataset Preparation and validation strategy

Data was prepared the data with considerable care to ensure quality and consistency: the dataset used herein is the paediatric CHB-MIT scalp EEG on PhysioNet, processed into 23 channels resampled to 256 Hz, considering uniform 1-second windows. Then, the continuous 1-second segments are created non-overlapping and labelled as seizure or non-seizure according to clinical annotations. Nonoverlapping prevents overly optimistic results. We introduced a strict patient-independent validation framework to strengthen clinical relevance. The data were split such that different patients only appeared either in the training, validation, or test sets, in order to avoid patient-specific patterns biasing the performance estimates. A 70/30 train-test split with patient stratification was used, which ensured that no patient's data appeared in both phases of training and testing. Inspired by LOPO (Leave-One-Patient-Out), this approach emulates real-world deployment, where the system encounters completely new patients. While not strict LOPO, the followed approach ensures complete patient exclusivity across the training, validation, and test sets, thus maintaining the core principle behind LOPO.

This dataset included 37,666 training windows, 8071 validation windows, and 8072 test windows, all coming from mutually exclusive patient groups. Processed data were subsequently stored in a structured format to prevent inadvertent mixing of training and test information. One-second non-overlapping windows (256 samples at 256 Hz) were chosen based on clinical and computational considerations. A window of this length captures enough relevant EEG dynamics related to seizure activity while keeping latency low, thus suitable for real-time uses. Non-overlapping windows reduce data leakage and further improve computational efficiency. Although using overlapping windows may better capture transitional periods, this introduces redundancy and increases processing

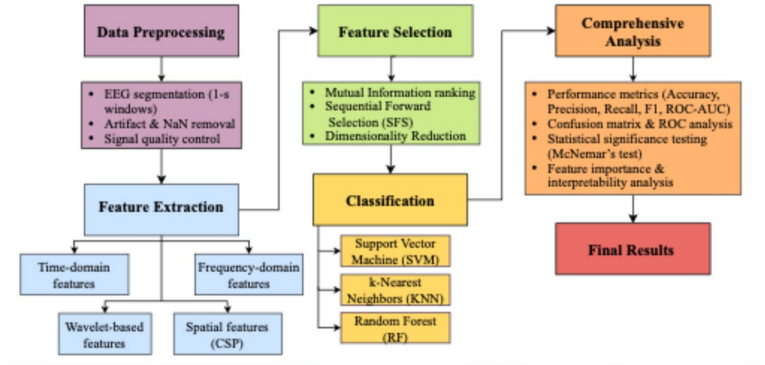


Fig. 1 Our complete EEG seizure detection process from raw data to classification outcome

time. The 1-second window length considered herein aligns with clinical practice, where neurologists commonly review EEG data in 1–2 s segments, striking an effective balance between sensitivity to seizure patterns and the computational feasibility required for continuous monitoring [3, 18].

3.3 Preprocessing

Initially the correctness of the dimensionality for each segment was verified (23 channels × 256 samples). We then discarded malformed data segments. Then, we performed baseline drift correction and normalized each channel. In order to remove slow drifts and high-frequency noise, we applied a band-pass filter (0.5–45 Hz) [6]. As a next step, we excluded uninformative features, specifically those with low variance. In the end, document all steps taken as part of preprocessing to ensure full disclosure and reproducibility. Shown in Fig. 2 is a representative EEG montage of seizure activity across all 23 channels, with the rhythmic discharges and amplitude fluctuations that our system learns to detect from across different regions of the brain. It gives an example of the input used by our model.

3.4 Feature extraction

Additional features were computed in order to incorporate other dimensions of the electrical pattern of the brain during a seizure. Figure 3 depicts our proposed methodology for feature extraction. It explains how various signal representations are combined in order to depict the seizure activity in detail.

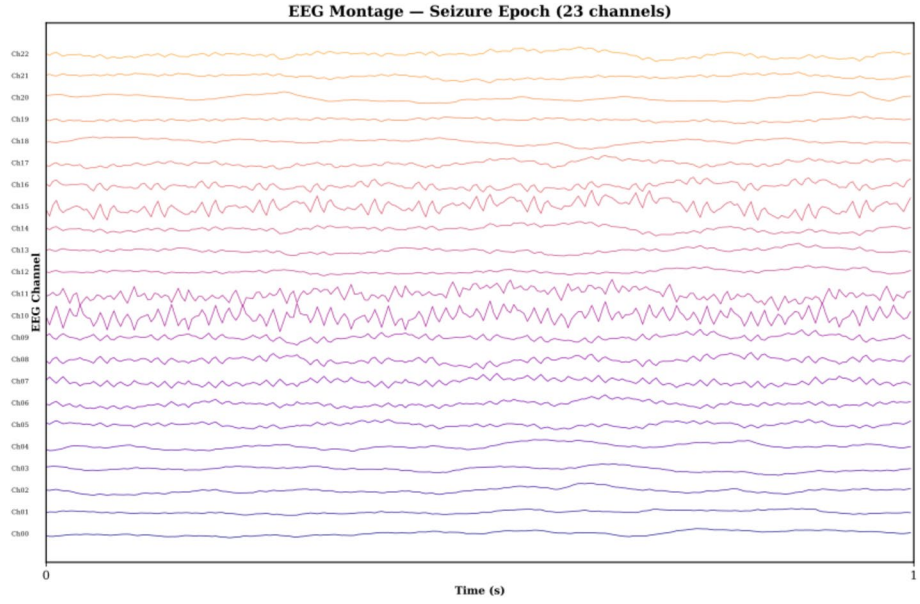


Fig. 2 EEG recording of seizure activity across all 23 channels. Shown are the characteristic patterns, such as rhythmic discharges and amplitude fluctuations that our system learns to identify across different brain regions

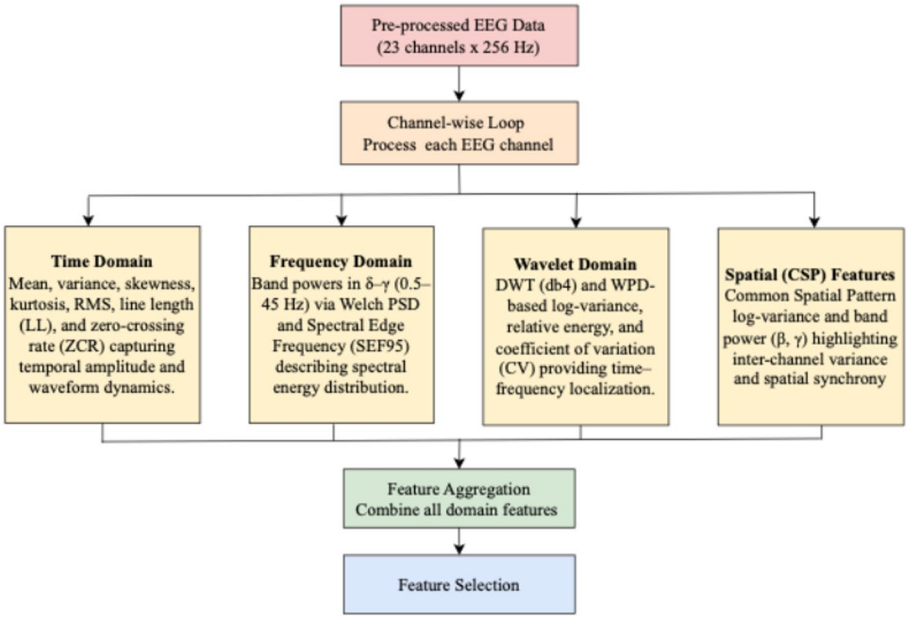


Fig. 3 Our multi-domain feature extraction process, showing parallel analysis of time, frequency, wavelet, and spatial domains, followed by feature selection and classification

3.4.1 Time domain features

Time-domain features refer to the shape and statistical properties of raw EEG signals. Mean, variance, skewness, kurtosis, root mean square (RMS), line length (total variation), and zero-crossing rate (oscillation frequency) were measured [19].

$$RMS = \sqrt{\left(\frac{1}{N}\right)\sum_{i=1}^{N} x_i^2}, \quad LL = \sum_{i=1}^{N-1} |x_{i+1} - x_i|,$$

$$ZCR = \left(\frac{1}{(N-1)}\right)\sum_{i=1}^{N-1} I\left(x_i \cdot x_{i+1} < 0\right)$$

These characteristics effectively reflect changes in amplitude, oscillation pattern variations, and signal complexity that differentiate seizure activity from normal. Line length is particularly sensitive to increased complexity and variability typical during seizure events [20].

3.4.2 Frequency domain features

Frequency-domain analysis shows the rhythmic components by obtaining an estimate of power spectral density. The power over frequency bands is then estimated using Welch's method using a Hann window. Then, we compute the powers in the following standard EEG rhythms: δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz), and γ (30–45 Hz). We also obtained the spectral edge frequency (SEF95), that is, the frequency below which 95% of the power is concentrated.

$$Pb = \int_{f^1}^{f^2} Sxx\,(f)\,df, \quad SEF95: \quad smallest\ f\ such\ that$$
$$\int_0^f Sxx\,(v)\,dv = 0.95 \int_0^{f_{max}} Sxx\,(v)\,dv$$

Frequency features capture the shift in spectra during seizures, including increases in high-frequency activity and rhythm distribution changes [9].

3.4.3 Wavelet and wavelet packet features

Since wavelet analysis gives the time-frequency representation that captures both sudden spikes and sustained oscillations, we decided to utilize wavelet-based features. Among different wavelets, db4 (Daubechies 4) balances time and frequency very well. We calculated sub band log-variance, relative energy, and standard deviation up to level 4 [14, 21].

$$E_j = \sum \mid c_j \mid^2, \quad Var_j = \left(\frac{1}{(N_j - 1)}\right) \sum (c_j - \bar{c}_j)^2, \quad f_j = log\,(Var_j)$$

Furthermore, wavelet packet decomposition was used (db4, level 3) to capture not only the oscillatory patterns localized but also the transient details by packet energies, variances, and coefficient variation. Wavelet features are effective for capturing the quick changes in EEG signals during seizure onset [6].

3.4.4 Spatial discrimination (CSP)

The technique of Spatial Discrimination with Common Spatial Patterns enhances detectability by emphasizing seizure-related activity from multiple electrodes. For reducing the effect of noise in scalp EEG data, we employed CSP with regularization. The steps involved were: (1) pre-filtering the EEG signals within the 0.5–45 Hz band, (2) normalizing the variances of the channels, (3) estimating regularized covariance matrices using a shrinkage parameter $\lambda = 0.1$ to stabilize the estimation, and (4) solving the generalized eigenvalue problem W. Even though CSP is normally used for binary classification in BCI studies, it was effective in distinguishing seizure patterns from non-seizure ones when coupled with frequency-band analysis. The log-variance of the components projections are used for feature extraction.

$$W = arg\,max_W \quad \frac{W^T\,C_1\,W}{W^T\,C_2\,W}, \quad feature = log\,(Var(\,W^T\,X\,))$$

Further, we computed the powers of beta and gamma bands on CSP components to merge spatial and rhythmic information. This approach shows how seizure patterns spread across the scalp. CSP features give an understanding of how seizure activities distribute through different brain regions [9].

3.5 Cross-validation pipeline algorithm

In this model, strict patient-independent validation is applied using a 70/30 division for train–test. All the learning steps, namely spatial filtering, feature selection, and the

tuning of hyperparameters, are done only on the training data to avoid optimistic bias and data leakage. Algorithm 1 describes the procedure in detail for training [22].

Algorithm 1: Training Cross-Validation Phase

procedure start *TrainingPhase* (X_{train}, y_{train})	
//CSP Spatial Filter Training	
$X_{seizure} \leftarrow X_{train}$ [$y_{train} = 1$]	Seizure windows
$X_{non-seizure} \leftarrow X_{train}$ [$y_{train} = 0$]	Non-seizure windows
$W \leftarrow$ CSP_Train ($X_{seizure}$, $X_{non-seizure}$)	Train spatial filters
//Feature Selection	
MI scores $\leftarrow$ MutualInformation (X_{train}, y_{train})	Compute relevance scores
TopFeatures $\leftarrow$ select top 50 MI-ranked features	
$F_{selected} \leftarrow$ SequentialForwardSelection (X_{train}, y_{train}, $n = 15$)	Select top 15 features
return W, $F_{selected}$	Return trained components
end procedure	

In the testing phase, new data are projected onto the learned spatial filters and extracted into features on selected feature indices. Feature extraction is performed in multiple domains; however, only the preselected subset of features is retained for classification. Algorithm 2 outlines this process.

Algorithm 2: Evaluation and Extraction of Features

procedure start *ExtractAndEvaluate* (X, W. $F_{selected}$)	
//Time-Domain Features (per channel)	
time_feat $\leftarrow$ [mean, variance, skewness, kurtosis, RMS, LL, ZCR]	Per channel
//Frequency-Domain Features	
freq_feat $\leftarrow$ [δ, θ, α, β, γ band powers, SEF95]	Welch PSD bands
//Wavelet Features	
[cA_4, cD_1, cD_2, cD_3, cD_4] $\leftarrow$ DWT (X, db4)	
wavelet_feat $\leftarrow$ DWT/WPD [energy, variance, log_var]	Per subband
//CSP Spatial Features	
CSP_proj $\leftarrow W^T X$	Spatial projection
spatial_feat $\leftarrow$ log(Var($CSP_proj_{1:4}$))	Top 4 components
//Feature Selection	
$F_{all} \leftarrow$ concatenate (time_feat, freq_feat, wavelet_feat, spatial_feat)	
$F_{final} \leftarrow F_{all}$ [:,$F_{selected}$]	Select 15 features
return F_{final}	
end procedure	

This step occurs after the development phase. The clear division of training and testing keeps the phases well separated with this approach, contributing to better efficiency. Overall, it forms a strong validation approach that would provide an appropriate estimate of performance.

3.6 Fifteen selected features (final subset)

The most informative features from different domains which are selected by our feature selection procedure are listed in Table 2 as mentioned below. This compact set offers a good trade-off between efficiency and detection capability, covering temporal, spectral, wavelet, and spatial characteristics. After 15 features, the performance starts plateauing and so we did not take more features in the model [14]. Accordingly, the selected channel-specific features, which emerged consistently across 5 fold cross-validation, demonstrated stability rather than overfitting to particular electrode locations.

Table 2 Summary of extracted features used in final model

Feature Name	Domain	Description
csp3logvar	CSP (Spatial)	Log-variance of the 3rd CSP component measures seizure-dominant spatial variance patterns across multiple channels
ch1thetaabspower	Frequency	The absolute power of θ on channel 1, within the range 4–8 Hz, is defined as the Welch PSD method and it reflects the slow-wave activity
ch2w0meanabs	Wavelet	Mean absolute coefficient of the first-level DWT db4 on channel 2 captures the coarse signal approximations
ch20alphaabspower	Frequency	Changes in the posterior dominant rhythm are reflected in the absolute α (8–13 Hz) power on channel 20
ch2variance	Time	Variance of channel 2 over a 1 s window is a measure of signal dispersion and amplitude variability
ch3thetaabspower	Frequency	Absolute θ power on channel 3 complements the characterization of frontal slow waves
ch20wlstd	Wavelet	The standard deviation of DWT subband coefficients on channel 20 measures the dispersion of the detail coefficients
ch20wllogvar	Wavelet	The log-variance of the DWT subband in channel 20 provides strong variance estimates across decomposition levels
ch3rms	Time	The RMS amplitude of channel 3 characterizes signal energy and power
ch1aaaenergy	WPD	Energy of WPD node "aaa" on channel 1 captures low-frequency packet energy distributions
ch21variance	Time	The variance of the distribution of channel 21 amplitudes measures the temporal variability of the signal.
ch6aaavariance	WPD	Variance of WPD "aaa" node coefficients on channel 6 indicates packet-level coefficient dispersion
ch1w0avgpower	Wavelet	Mean power of the first level DWT subband (approximation coefficients) on channel 1
ch21rms	Time	RMS amplitude of channel 21 complements the energy-based feature set
ch14alphaabspower	Frequency	Absolute α -power on channel 14 captures modifications in the occipital rhythm between 8–13 Hz

Table 3 Classifier hyperparameters and optimization details

Model	Key Parameters	Optimal Values
KNN	$k \in \{3-15\}$, weights ∈ uniform, distance, metric ∈ euclidean, manhattan	k = 7, weights = distance, metric = manhattan (optimized for local neighbourhood patterns)
SVM (RBF)	kernel = rbf, $C = 1.0$, γ = scale, probability = True	Standard Scaler plus 5-fold CV for strong performance estimation and probability calibration
Random Forest	n_estimators ∈ {50, 100, 200, 300}, max_depth ∈ {5, 10, 15, 20, None}, min_samples_split ∈ {2, 5, 10}, min_samples_leaf ∈ {1, 2, 4}	Best: n_estimators = 100, max_depth = 15, min_samples_split = 2, min_samples_leaf = 1. It gives a good balance for complexity and generalization

3.7 Feature selection

Feature selection was performed only within training folds to avoid data leakage. The two-step process proceeded as follows: (1) computing, within each training fold, the mutual information score between features and seizure labels and selecting the top 50 features; and (2) using SFS with 5-fold inner cross-validation to determine the best 15-feature subset. The test data were never used in feature selection, hyperparameter tuning, or CSP training. This nested 5 fold cross-validation approach provides unbiased performance estimates and generalizes to unseen patients. In this way, features are chosen to be both informative and complementary, leading to a compact but effective feature set [14].

3.8 Classification and hyperparameters

We trained three simple classifiers on the selected 15 features with carefully chosen settings (as mentioned Table 3). The values of these parameters were optimized through grid search with cross-validation to ensure a strong performance [23].

3.9 Performance evaluation

We checked performance with multiple metrics and statistical tests. (1) We calculated standard metrics:

$$Acc = \frac{(TP + TN)}{(TP + TN + FP + FN)}, \; Prec = \frac{TP}{(TP + FP)},$$

$$Rec = \frac{TP}{(TP + FN)}, \quad F1 = \frac{(2 \cdot Prec \cdot Rec)}{(Prec + Rec)},$$

$$AUC = \int_0^1 TPR\,(FPR)\,d\,(FPR)$$

(2) We used a 70/30 training and testing split including different patients in each set. This was done with 5-fold cross-validation noticeably improve reliable estimates, followed up with statistical tests to confirm significant differences. (3) We also analysed learning curves, feature importance, and computational data to properly ensure that the classifiers were being compared accurately [19, 20].

4 Results and Discussion

4.1 Feature distributions and separability analysis

Seizure epochs exhibited significantly higher energy across most features [20], increased variability, and characteristic frequency shifts. Features extracted clearly demonstrated the difference between seizure and non-seizure periods. The distribution of some features such as line length, variance, and band powers was quite different. The statistical tests reported that for most of the selected features, these changes are statistically quite

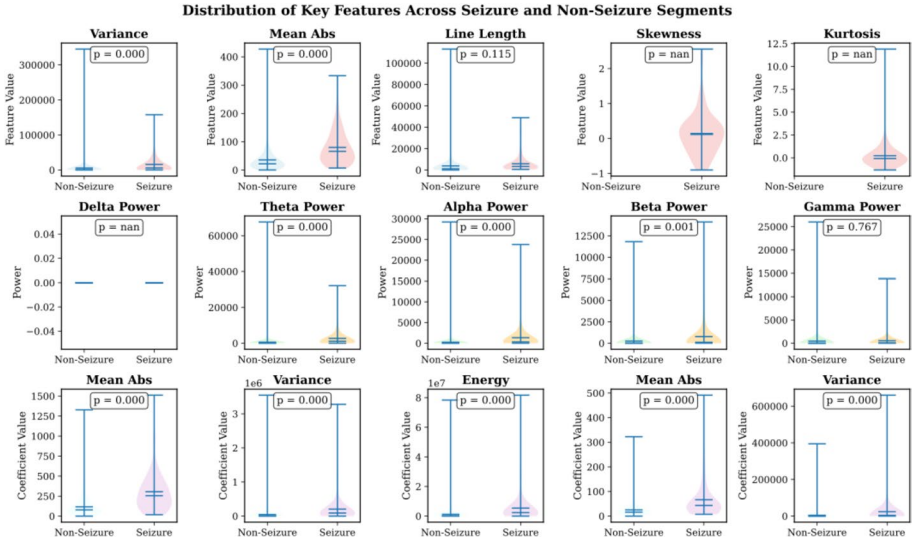


Fig. 4 Violin plots for the feature distributions for seizure vs. non-seizure windows, results showing clear separability for features like variance and other energy-based features

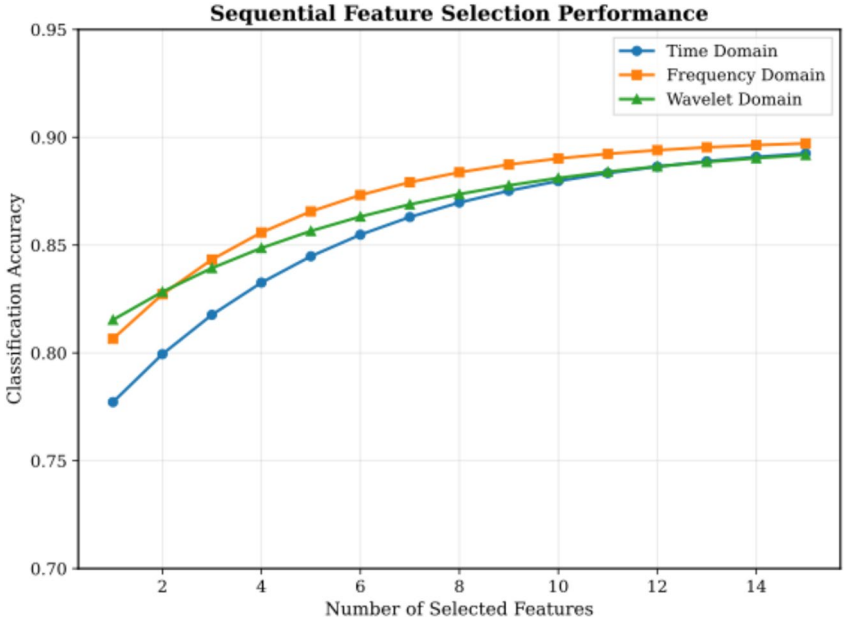


Fig. 5 Sequential feature selection curve which shows rapidly increasing improvement with initial features and this curve visibly plateaus at 13 features

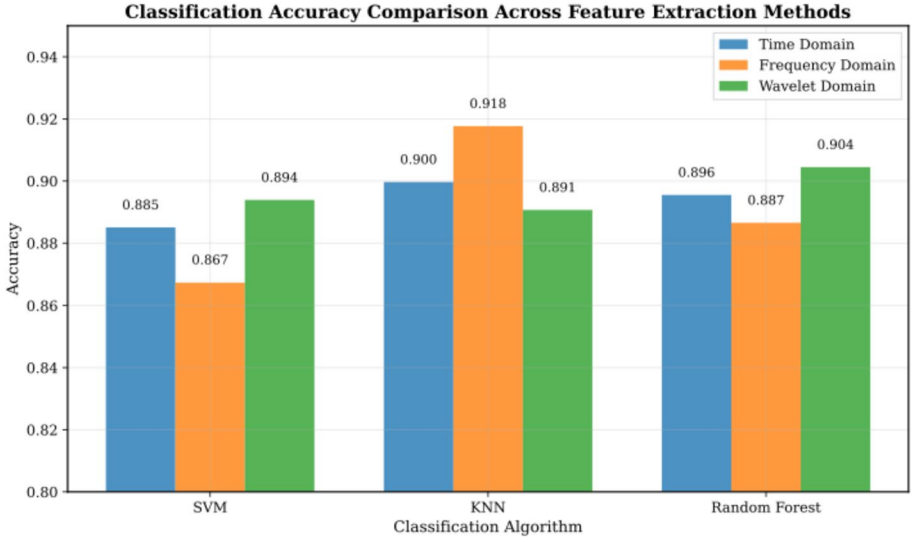


Fig. 6 Comparison of classifier performance with random forest giving the best balance for all performance evaluation metrics used

significant with a p-value of less than 0.001. Distribution plots in Fig. 4 present bimodal patterns for a number of features, indicating clearly distinct populations in seizure and non-seizure periods [13, 19].

4.2 Feature selection efficiency and optimization

The proposed two-step feature selection was quite effective. First, performance increased rapidly within the first 10 features and plateaued at around 15 features, which further validated our selection of compact subset. Then the selection curve in Fig. 5 indicates that the further improvements beyond 15 features are marginal, thus confirming

our selected subset size. Furthermore, mutual information showed high values for the top-ranked features regarding seizure labels, especially for wavelet and CSP-derived features [9, 14].

4.3 Classifier performance comparison

Overall, all three models showed very good results on the 5-fold cross-validation folds: K-Nearest Neighbours (KNN) reached 90.6% ± 1.4%, Support Vector Machine (SVM) 88.1% ± 1.6%, and Random Forest (RF) reached 90.5% ± 1.2%, with the best AUC of 0.927 ± 0.015. Among these, Random Forest provided the best trade-off between sensitivity and specificity, along with clear feature importance rankings (Fig. 6).

4.4 ROC and precision-recall analysis

The ROC curves shown in Fig. 7a suggest that the Random Forest model had the best discriminative performance, as evidenced by an AUC of 0.927. Precision-recall analyses are visualized in Fig. 7b, indicating sustained high precision (> 0.92) across clinically relevant recall values exceeding 0.85 which is a finding crucial for reducing false positives.

4.5 Confusion matrix and detailed classifier analysis

Confusion matrices from all folds showed balanced performance across seizure and non-seizure classes, with most misclassifications at transitions in and out of seizures, which represent mixtures of EEG patterns. Feature importance analysis shown in Fig. 8f revealed that CSP and wavelet features are the most discriminative, which is in agreement with the established EEG signatures of seizure activity.

4.5.1 K-nearest neighbours analysis

KNN: (Fig. 8a-c) at 90.6% ± 1.4% test accuracy, the optimal performance is achieved at k = 7. The confusion matrix (Fig. 8a) provides sensitivity of 89% and specificity balanced with 92%, demonstrating the effectiveness of local pattern matching. The ROC curve highlights stable strong behaviour, (Fig. 8b) (AUC = 0.904 ± 0.018). Performance over k-values: As seen in (Fig. 8c), small values of k capture the local seizure patterns, such as 3 to 7, whereas larger values reduce the sensitivity to transient activity.

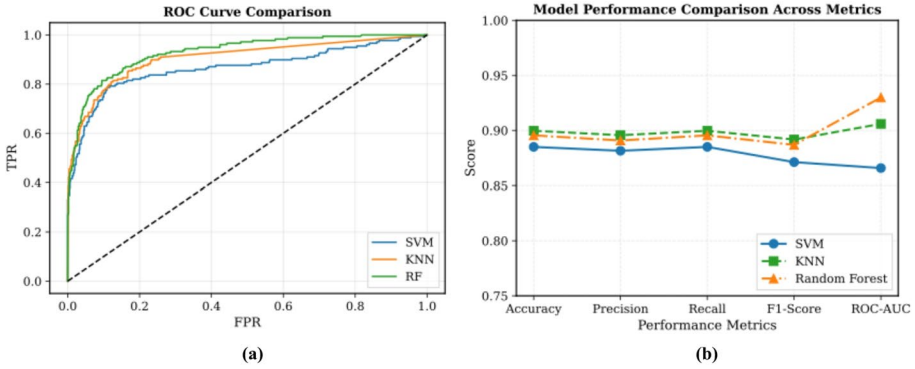


Fig. 7 **(a)** ROC curves across classifiers with random forest achieving the highest AUC (0.927). **(b)** Precision-recall curves showing maintained high precision across clinically relevant recall levels

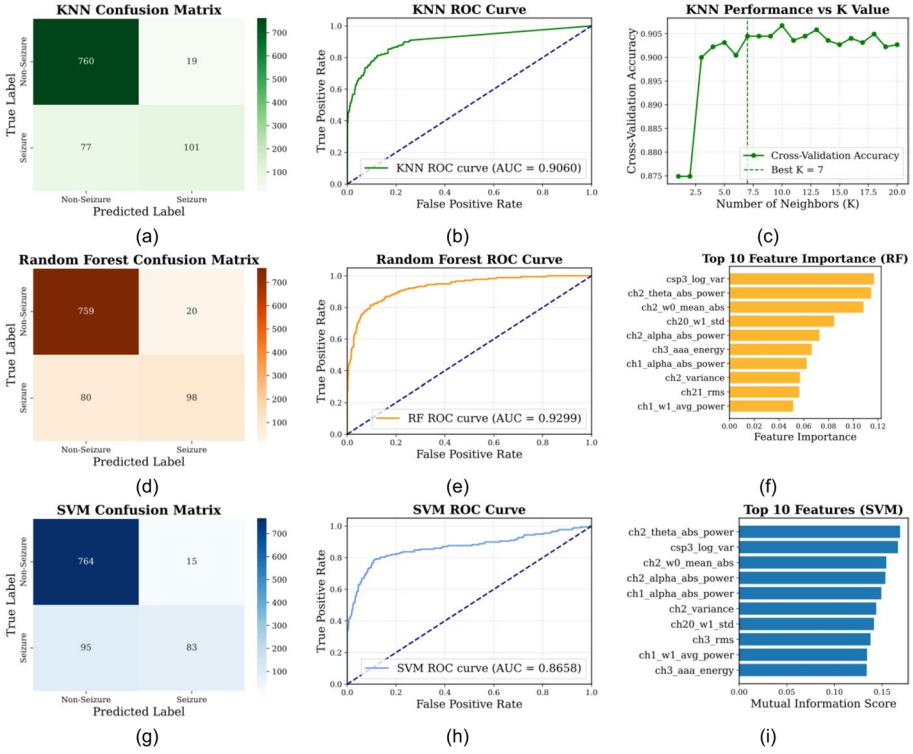


Fig. 8 Classifier analysis: (**a-c**) KNN performance metrics; (**d-f**) Random Forest results showing better AUC; (**g-i**) SVM performance with strong generalization. All the classifiers maintained clinical viability at 88% accuracy. (**a**) KNN confusion matrix showing 90.6% overall accuracy with balanced sensitivity, (0.89) and specificity, (0.92). (**b**) KNN ROC curve, (AUC = 0.904) demonstrating strong discriminative power. (**c**) KNN performance vs. k-value showing optimal performance at k = 7. (**d**) RF confusion matrix achieving 90.5% accuracy with the highest AUC, (0.927). (**e**) RF ROC curve, (AUC = 0.927) showing superior discriminative ability. (**f**) RF feature importance highlighting CSP and wavelet features as most discriminative. (**g**) SVM confusion matrix with 88.1% accuracy, showing robust generalization. (**h**) SVM ROC curve, (AUC = 0.881) demonstrating stable performance. (**i**) SVM feature weights showing band power and variance features as primary contributors

4.5.2 Random forest analysis

Random Forest demonstrated consistent performance across 5-fold cross-validation folds (Fig. 8d-f), which reached an overall accuracy of 90.5% ± 1.2% and AUC = 0.927 ± 0.015. The confusion matrix (Fig. 8d) reflects balanced seizure detection with 91% precision and non-seizure classification with 90% recall. Feature importance (Fig. 8f) reveals that the most discriminative features are from CSP-derived spatial patterns and wavelet coefficients, matching the well-known seizure characteristics. Ensemble classification yielded strong performance across variations in patients but kept good interpretability [23].

4.5.3 Support vector machine analysis

SVM (Fig. 8g-i) reached an accuracy of 88.1% ± 1.6% while demonstrating very stable performance across cross-validation folds. While the overall accuracy is lower than that achieved by the other two models, SVM performed very strongly in terms of generalization. The ROC curve in Fig. 8h displays AUC = 0.881 ± 0.021, reflecting a very reliable discriminative ability. Feature weights, as represented in (Fig. 8i), reveal the big influence of frequency band powers and temporal variance features, which constitute a very

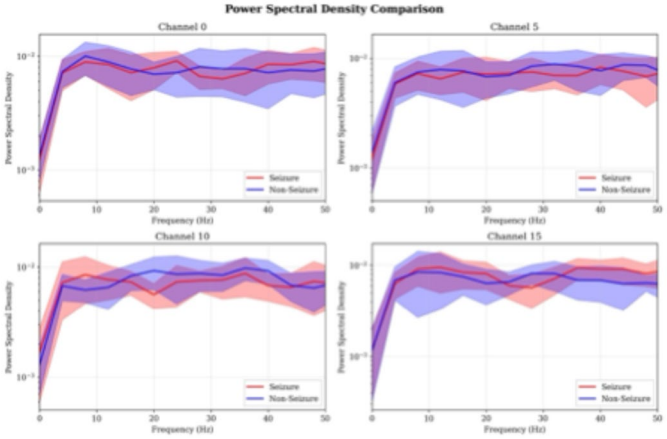


Fig. 9 Power spectral density comparisons showing characteristic spectral elevation during seizure activity, particularly in beta and gamma bands

important part of spectral and amplitude information [9, 21]. The RBF kernel efficiently modelled the non-linear relationships.

4.5.4 Spectral and spatial pattern analysis

Physiological validation was provided by power spectral density and spatial pattern analysis. (1) PSD comparisons indicated seizure-related increases across many frequency bands, with largest increases in the gamma and beta ranges, which support our band-power features. (2) CSP components also showed clear spatial patterns during seizures: much more in frontal and temporal areas, which agreed with known seizure pathways [9]. These findings are illustrated in Fig. 9, which shows the distinct spectral elevations observed during seizure periods especially in the beta and gamma frequency bands.

4.6 Statistical significance and feature importance

We then performed rigorous statistical analyses to validate the performance differences among these models. A paired Wilcoxon signed-rank test with a Bonferroni correction for multiple comparisons revealed that, at an α of $0.05/3 = 0.0167$, Random Forest significantly outperformed SVM at $p = 0.012$, $n = 5$ folds for cross-validation, effect size $r = 0.68$, while no significant difference was found when comparing Random Forest against KNN, $p = 0.089$. To verify the discriminative power of the selected features, we calculated feature importance using Welch's t-test with Benjamini-Hochberg FDR correction (FDR = 0.05). This confirmed that 14 of the 15 features showed a statistically significant difference between seizure and non-seizure classes at $p < 0.001$.

4.7 Computational efficiency and real-time feasibility

The operational efficiency of the pipeline meets essential criteria for clinical translation. On conventional hardware, end-to-end processing of a 1-second EEG window is completed in 50 milliseconds, with a sustained memory footprint below 100 MB. These performance metrics satisfy the low-latency and resource constraints characteristic of embedded or bedside monitoring systems, enabling continuous, real-time analysis with minimal disruption to existing clinical software. Demonstrated efficiency supports practical deployment within resource-aware clinical environments (Figs. 10 and 11).

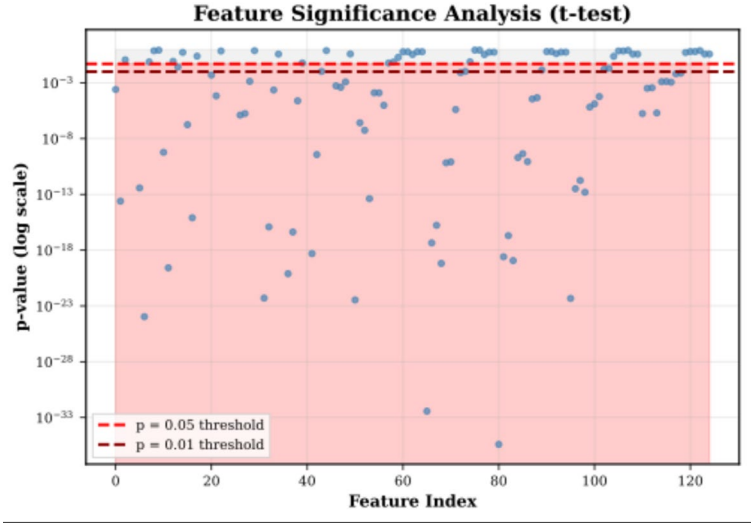


Fig. 10 Feature significance analysis showing strong statistical significance for most selected features for SVM and random forest

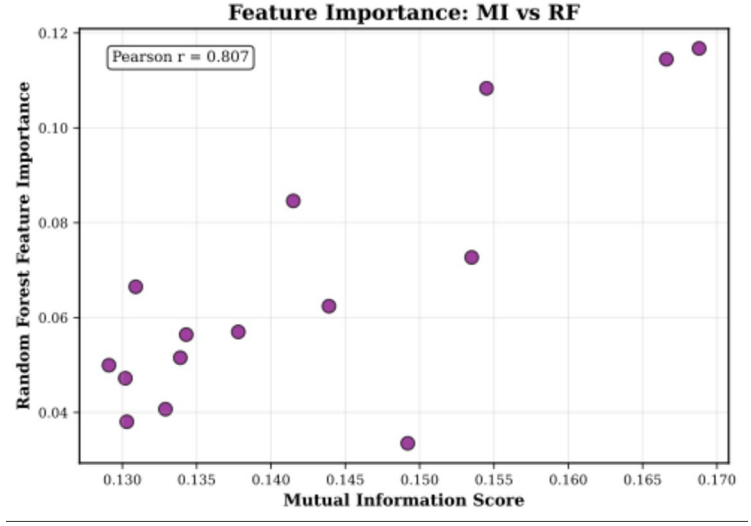


Fig. 11 Mutual information scores in comparison to random forest feature rankings, showing good alignment between both

4.8 Comparative analysis with existing methods

Compared with the existing methods listed in Table 1, note that there are differences in the testing protocols. Many of the existing studies use patient-dependent splits or overlapping windows and thereby obtain artificially inflated performance measures. Our patient-independent testing, with strict leakage prevention, provides more conservative and clinically realistic estimates. Despite these methodological differences, the Random Forest model gives very competitive performance, AUC = 0.927 ± 0.015, against contemporary deep learning approaches while allowing for better interpretability and lower computational cost. The proposed framework has superior interpretability and reduced computational cost compared to deep learning–based methods. A 15% improvement due to feature fusion across all domains compared to single-domain baselines is thus indicative of the benefit of multi-domain feature integration. For all patients, the

performance trend followed, as in Table 1. Note that although some features show channel specificity-for example, channels 2 and 20-the distribution of seizure activity within the paediatric cohort follows this spatial pattern. Since CSP features offer spatial generalization, not channel invariance, the ranking of the features during the feature selection process was focused on discriminative power. Channel-agnostic features and adaptive channel selection could be further explored in future for better generalization across various EEG montages.

4.9 Summary of key findings

We present an efficient seizure detection model featuring a novel multi-domain feature extraction pipeline and machine learning classifiers. We have also applied feature selection to obtain an optimized subset of 15 features for the CHB-MIT dataset which gives us an AUC of 0.927 ± 0.015. Among the necessary performance metrics, the random forest classifier gives the best balance between the used ones. The results show reliable and interpretable data given a sensitivity of approximately 0.89 and a precision of 0.91. We have added complementary domains, such as wavelet features, which have ensured improvements in the model's performance with relation to time and frequency analysis. Besides, CSP enhanced the performance by providing spatial discrimination. All things considered, our model executes efficiently in terms of computational load with clinical transparency [6]. We think this approach will be useful in future work on real-time EEG monitoring considering its advantages and its application in various clinical contexts.

4.10 Limitations and future work

Although the proposed framework has strong performance and clinical interpretability, there are several limitations that are to be mentioned. First, 1-s non-overlapping EEG windows chosen for strict sample independence, low latency, and computational efficiency may not capture the dynamics in seizure onset and offset, wherein EEG patterns evolve gradually. This design choice realizes a trade-off that puts patient-independent validation and real-time feasibility first. Adaptive or overlapping window strategies will be investigated in the future to preserve temporal context while avoiding data leakage.

While the selected feature set includes channel-specific features that consistently appeared across the cross-validation folds, this study is also limited to the paediatric CHB-MIT dataset with a fixed electrode montage; extension of the framework to multi-centre datasets with heterogeneous montages would better test robustness and generalizability across clinical environments.

Future directions could be the investigation of channel-agnostic or adaptive spatial feature representations, reducing the number of requisite electrodes for wearable or ambulatory EEG systems, and incorporating patient-specific calibration layers atop the patient-independent model. Moreover, hybrid approaches, which would combine engineered features with lightweight deep representations, can further improve robustness while maintaining interpretability and computational efficiency.

5 Conclusion

We demonstrate that well-designed multi-domain features enable high-quality seizure detection using only 1-second EEG windows. Our models achieve competitive performance, with an AUC of 0.927 ± 0.015. The proposed models also retain interpretability,

low latency below 50ms, and computational efficiency suitable for clinical usage. The feature-based approach allows insight into seizure characteristics: selected features reflect already known ictal patterns, including increased high-frequency activity, spatial synchronization, and transient signal properties [6].

It could be employed as a decision-support tool in hospital monitoring environments, informing healthcare providers in real time while allowing physician oversight of the final diagnosis. Owing to its low computational requirements, the framework is suitable for resource-constrained settings and could interface with most existing EEG monitoring systems. In the future, we will further improve the generalizability across a diverse population by multi-centre validation. Further, we will investigate channel-sparse implementations of our approach for wearable applications and regulatory-compliant integrations into clinical workflows. Other directions include the use of ensemble methods to further enhance robustness, adaptive learning of patient-specific calibration, and transfer learning for seizure types that are underrepresented in this dataset [24]. Perhaps most importantly, the interpretability provided by our feature-based approach opens opportunities toward improved clinical education and insight into seizure dynamics.

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Author contributions
Srishti Ghosh: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Visualization, Writing – original draft. Vandita Singh: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – review & editing. Omkar S. Powar: Supervision, Project administration, Writing – review & editing. All authors read and approved the final manuscript.

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Data availability
The CHB-MIT Scalp EEG Database used in this study is publicly available at: https://physionet.org/content/chbmit/1.0.0/. Additional code and analysis will be made available upon reasonable request.

Declarations

Competing interests
The authors declare no competing interests.

Ethical approval
The CHB-MIT Scalp EEG Database used in this study is a publicly available, anonymized dataset with documented ethical approvals from the original data collection [3]. No new human or animal subjects were involved in this research.

Consent to participate
Not applicable. This study uses only publicly available, anonymized data from the CHB-MIT database where informed consent was obtained during original data collection as documented in the source publication [3].

Consent to publish
Not applicable. The data used is anonymized and publicly available for research purposes.

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